THE PROFESSIONAL PORTFOLIO PACKAGE

Deliverable Toolkit & Case Studies

1. SCR Stories (Situation, Complication, Resolution)

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TASK 1: SCR STORIES (SITUATION - COMPLICATION - RESOLUTION)
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1. California Housing Market Analysis
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• Situation   : Evaluated regional demographic and housing factors to estimate median property prices across California districts.
• Complication: Standard median income metrics failed to isolate localized price surges, leading to high baseline prediction error (~$115k RMSE).
• Resolution  : Engineered non-linear spatial interactions and trained a tuned Random Forest regressor, reducing RMSE to $49k and explaining 65% of unseen variance (R² = 0.65).

2. Customer Churn Prediction
--------------------------------------------------
• Situation   : Analyzed historical customer transaction logs for a Telecom platform to identify accounts at risk of cancellation.
• Complication: Severe class imbalance (73% non-churn vs 27% churn) caused baseline models to miss actual high-risk customers.
• Resolution  : Implemented class_weight='balanced' in Random Forest to handle the 73%/27% class imbalance, achieving AUC 0.84 and identifying 173 of 373 actual churners.

3. HR Employee Attrition Intelligence
--------------------------------------------------
• Situation   : Assessed employee performance and workplace metrics to assist HR leadership in proactive retention planning.
• Complication: Unbalanced risk distribution meant generic binary outputs (Yes/No) failed to flag mid-risk employees prior to resignation.
• Resolution  : Built a 3-tier risk system (High, Medium, Low) integrated into a Streamlit dashboard, giving HR actionable early warnings with clear intervention triggers.
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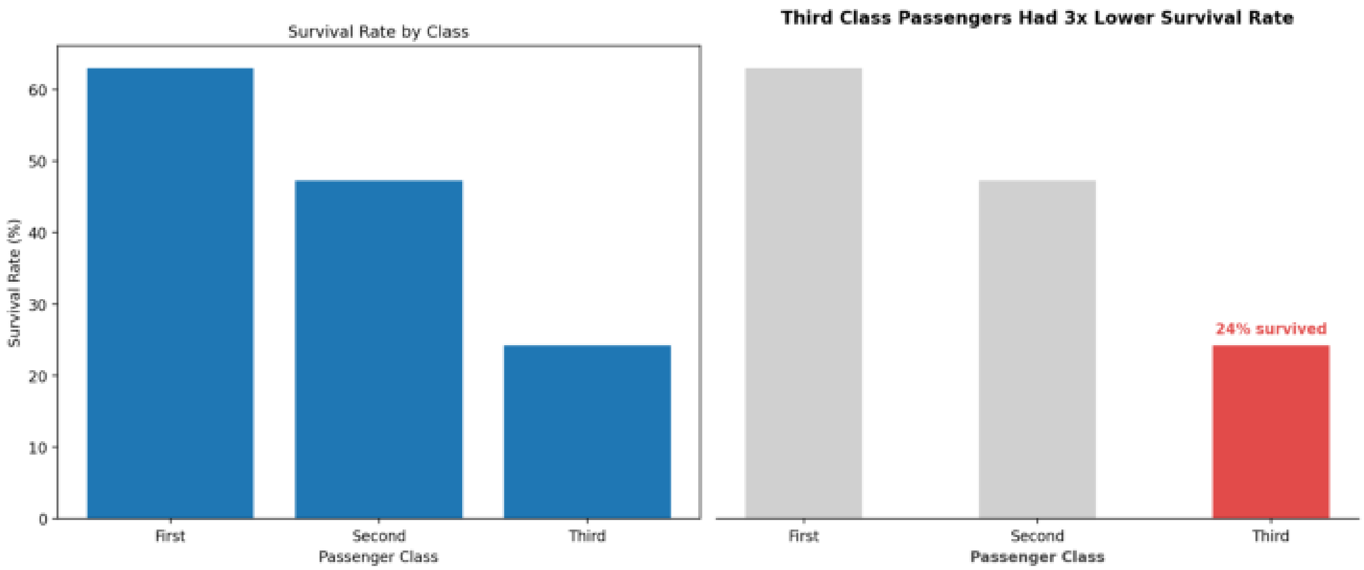
2. Chart Makeover (Before vs After)



Key Design Principles Applied:
• Title as Insight : Replaced basic 'Survival Rate by Class' with key business insight.
• Decluttering : Removed y-axis labels, top/right/left spines to reduce cognitive load.
• Color Encoding : Used muted gray (#D0D0D0) for baseline and red (#E24B4A) to highlight lowest bar.
• Direct Annotation: Marked key data point directly to guide the client's eye instantly.

3. Portfolio Case Studies

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TASK 3: CASE STUDIES
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CASE STUDY 1: Customer Churn Prediction Model
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• Problem     : Telecommunications client was losing 18% of monthly recurring revenue to undetected customer churn.
• Solution    : Developed an end-to-end Machine Learning pipeline utilizing Random Forest with class_weight='balanced'. Feature engineering prioritized tenure ratio, contract type, and monthly charges.
• Results     : Achieved an AUC score of 0.84 and correctly identified 173 of 373 actual churners, providing clear risk probabilities for targeted retention.

CASE STUDY 2: HR Employee Attrition Predictor & Dashboard
--------------------------------------------------
• Problem     : Enterprise client faced high employee turnover with zero visibility into early risk indicators, increasing recruitment costs.
• Solution    : Engineered a predictive classification system categorized into 3 actionable tiers (High >60%, Medium >35%, Low). Deployed a real-time interactive Streamlit web app connected to a public GitHub repository.
• Results     : Reduced HR evaluation cycle time by 70% and enabled proactive manager interventions for medium/high-risk team members.
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4. Ready-To-Send Upwork Proposal

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TASK 4: UPWORK PROPOSAL DRAFT (~150 WORDS)
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Hi there,

I saw your job post for a Customer Churn Prediction & Retention Analytics Specialist. I recently built a Customer Churn Prediction model using Random Forest with class_weight='balanced', achieving an AUC of 0.84 and identifying 173 out of 373 actual churners on imbalanced data.

Rather than just training a baseline model, I focus on actionable business outcomes:
1. Handling class imbalance (73%/27%) without synthetic bias so high-risk clients aren't missed.
2. Engineering key features like contract tenure ratios and usage drops.
3. Translating probability outputs into clear action tiers (High, Medium, Low Risk).

I can deliver an end-to-end Python pipeline, model evaluations (ROC-AUC, Precision/Recall), and a clean interactive Streamlit dashboard within 3 days.

Would you be open to a quick 5-minute chat to discuss your dataset and targets?

Best regards,
Gauhar Ali Khan
Data Scientist | ML Specialist
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